1.10.6 Constant Function

A function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = c,\ \forall\ x \in \mathbb{R}$ is called a constant function (Fig.1.50).

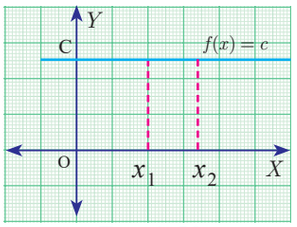


Fig. 1.50

Progress Check

1. Is a constant function a linear function?
2. Is quadratic function a one – one function?
3. Is cubic function a one – one function?
4. Is the reciprocal function a bijection?
5. If $f : A \to B$ is a constant function, then the range of f will have _____ elements.

Exercise 1.5

1. Using the functions f and g given below, find $f \circ g$ and $g \circ f$. Check whether $f \circ g = g \circ f$.

   (i) $f(x) = x - 6,\ g(x) = x^2$ (ii) $f(x) = \frac{2}{x},\ g(x) = 2x^2 - 1$

   (iii) $f(x) = \frac{x+6}{3},\ g(x) = 3 - x$ (iv) $f(x) = 3 + x,\ g(x) = x - 4$

   (v) $f(x) = 4x^2 - 1,\ g(x) = 1 + x$

2. Find the value of k, such that $f \circ g = g \circ f$

   (i) $f(x) = 3x + 2,\ g(x) = 6x - k$ (ii) $f(x) = 2x - k,\ g(x) = 4x + 5$

3. If $f(x) = 2x - 1,\ g(x) = \frac{x+1}{2}$, show that $f \circ g = g \circ f = x$

4. If $f(x) = x^2 - 1,\ g(x) = x - 2$ find a, if $g \circ f(a) = 1$.

5. Let $A, B, C \subseteq \mathbb{N}$ and a function $f : A \to B$ be defined by $f(x) = 2x + 1$ and $g : B \to C$ be defined by $g(x) = x^2$. Find the range of $f \circ g$ and $g \circ f$.

6. Let $f(x) = x^2 - 1$. Find (i) $f \circ f$ (ii) $f \circ f \circ f$

7. If $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are defined by $f(x) = x^5$ and $g(x) = x^4$ then check if f, g are one-one and $f \circ g$ is one-one?

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